

Dream-RSI Methods and Results

A construction log for long, genuinely readable English letter-level palindromes | Snapshot: 2026-09-18

Current status: no reader-worthy output has cleared the full gate yet. This report is intentionally not a negative-result thesis: it records the working search machinery, the exact diagnostic surfaces, and the next geometry-changing repair required to reach a long readable result.

1. Acceptance gate

A candidate is admitted only when (1) normalized letters equal their reverse, verified by an independent two-pointer check and an independent forward/reverse SHA-256 check; (2) provenance shows fresh construction rather than catalogue text, finished-tape reversal, word-order symmetry, repeated units, or self-palindromic words/spans; (3) the surface is intact ordinary English prose; and (4) blinded human readers prefer the intact passage to a randomized word-shuffled control. Programmatic language measures may filter or diagnose, but never certify readability.

2. Dream-RSI controller

Each completed construction lane writes a replay node: rendered surface, normalized tape, exactness hashes, mismatch mask, grammar and semantic state, provenance flags, and a concrete next repair. Replay selects a new operator on a held-out split; a fresh branch then executes that operator. A failure is progress only when it changes the construction geometry and names the next repair. The controller is an orchestration layer, never a readability judge.

The current live operator freezes matched outer character assignments, reopens the seam-owning typed constituents, and lets lexical content, inflection, and word boundaries change while carrying the mirrored character obligation immediately. The next branch is a single discourse grammar: a complete question plus a complete answer clause generated jointly across the seam, with all proper palindromic subspans forbidden.

3. Actual Dream-RSI results

Round / lane	Fresh work	Exact / admitted	Reader status
Model-guided span resynthesis	12 local GPT-2 seam proposals; exact rows were constructed by mirroring the residual; 86-120 letters.	12 exact by construction / 0 shortcut-free	Human-unreviewed; reflected residual had no lexical segmentation.
Replay round 119	21,990 independently audited nodes; 120 exact-but-fragment controls entered replay.	0 admissible exact	No reader gate; held-out mismatch-first stayed 0.780.
Dual-boundary model authoring	8 left x 8 right model proposals; 64 fresh paired worlds; no seed scaffold in output.	0 exact; longest 58 letters	Human-unreviewed; next repair reopens a complete residual word.
Discourse ellipsis + cross-word seam	148 exact diagnostic rows, 42-46 letters; 100 passed word-order check, but all retained rows have a forbidden proper palindromic island.	148 exact / 0 strict	Reader gate closed; next repair jointly authors a complete answer clause.

The strongest exact diagnostic surface

Independent audit: 44 normalized letters; two-pointer exact = true; forward SHA-256 = reverse SHA-256 = 0419d58b6773dcca6d527271d41fd0197da9dde876598c9dadfc87fa4f1a9f83. It is not admitted: the surface contains a proper palindromic island and therefore fails the shortcut-free gate. It has not been sent to readers.

Was Noel an era, a gas, an item? Met in a, saga, arena, Leon saw.

This row is useful because it demonstrates genuine cross-word boundary resegmentation (an era / arena, a gas / saga, an item / met in a) while also exposing the remaining failure mode: an exact tape can still be an engineered word-mirror or contain a proper palindrome island. The construction must now generate the whole discourse clause jointly and reject every such subspan before closure.

4. Fifty distinct construction methods and results

The matrix below is extracted from the novelty registry. Each row is a distinct construction signature, not a repeated parameter sweep. Results preserve the registry's wording; entries marked preflight have no run result and are not evidence of success. Exact diagnostic rows remain quarantined unless every shortcut-free gate passes.

#	Method / operator	Construction signature	Observed result and next repair
1	semordnilap crossword resegmentation	fresh-scene-left-arm / independent-semordnilap-word-pairs / cross-word-boundary-shifts / typed-clause-realization / two-pointer-sha-audit / no-repeated-uni	diagnostic run; 150 crossword boundary probes, 0 exact closures, longest 55 letters; independent SHA audit rejects all rows; next: first-residual typed lexical expansion
2	cross attachment resegmentation	cross-attachment-character-carry / determiner-adjunct-resegmentation / typed-valency / independent-two-pointer-hash-audit	diagnostic run; 0 equation nodes and 0 exact closures; intact controls 58-70 letters; reader gate closed; next: typed adjunct insertion with boundary carry
3	typed morpheme compound boundary	typed-prefix-stem-suffix / compound-boundary / independent-ordinary-clauses / inflectional-derivational-variants / independent-pointer-sha / no-word-mirror	diagnostic run; 54 probes, no exact closure, longest 59 letters
4	adjunct boundary targeted repair	single-adjunct-boundary-repair / fresh-preposition / preserved-typed-svo / live-seam-residual / independent-audit	repair run; exact closure unresolved
5	fresh typed frame live seam	fresh-typed-svo-frame / agreement-preserved / live-mirrored-character-obligation / seam-residual-ledger / independent-audit	repair run; residual obligations remain
6	clause pair csp central pivot	typed-svo-valency / independent-scene-pairs / central-pivot / character-tape-csp / no-word-mirror	diagnostic run; 113-letter intact prose, unresolved character debt
7	semantic slot lattice smt	semantic-slot-lattice / global-character-equation-pruning / joint-left-right-selection / weighted-residual-repair / independent-audit	diagnostic run; 108 rendered states, 0 accepted
8	bidirectional scene decoder	bidirectional-character-lm / live-semantic-scene-slots / outside-in-tape / independent-audit	diagnostic run; exact catalogue palindrome rejected
9	wordpair graph	independent-word-pair-graph / pos-valency-state / distinct-word-pairs / character-tape-closure / independent-pointer-hash	diagnostic run; 164-letter intact prose frontier, 0 exact closures
10	paired semantic mutation	quarantined-control / paired-mirrored-lexical-span / scene-slot-grammar / live-character-obligations	diagnostic run; 2 rendered candidates, 0 exact and 0 admitted
11	corpus backed reverse segmentation	fresh-clause-tape / weighted-word-boundary-dp / independent-pointer-hash / copied-span-rejection	diagnostic run; 2 fresh clauses, 0 exact closures
12	seam feature slot repair	targeted-seam-feature-slot-repair / single-lexical-substitution / live-reflected-seam / independent-exact-audit	repair run; substitutions did not close exact seam
13	typed boundary resegment shortwords	typed-first-failing-seam / boundary-resegmentation / adjacent-short-word-absorption / fresh-prose-control / independent-pointer-sha-audit	repair run; 23 variants, 0 exact and 0 mechanically admitted
14	char lm tape resegment	character-trigram-constrained-decoding / reverse-tape-dynamic-programming-resegmentation / ordinary-prose-seeds / full-tape-replay / independent-hash-audit	run; 1 exact closure rejected as catalogue/self-unit and 4 non-exact prose proposals
15	constructive seam morph cfg	dependency-tree-seam-csp / agreement-carrying-morphology-transducer / cfg-earley-character-intersection / live-tape-construction / independent-exact-audit	exact witness: Ava saw radar was Ava

#	Method / operator	Construction signature	Observed result and next repair
16	centerout paired boundary csp verbframe	paired-subject-adjunct-boundary-csp / fresh-verb-frame / ordinary-svo-grammar / live-character-obligation / independent-exact-hash-audit	run; 0 exact closures and 0 mechanically admitted probes
17	free center semantic state machine	free-center-pivot / role-agreement-state / live-bilateral-character-debt / complete-clause-growth / duplicate-sweep-rejection / independent-pointer-sha	diagnostic run; 138-letter intact prose, unresolved character debt
18	reverse lexicon adjacent pp boundary	reverse-lexicon-role-noun-boundary / adjacent-prepositional-attachment / two-boundary-residual-decrease / live-reverse-chart / heldout-scene / independent-	repair run; 0 exact closures and 0 mechanically admitted probes
19	centerout paired boundary csp	paired-subject-adjunct-boundary-csp / ordinary-svo-grammar / live-character-obligation / event-preserving-valency / bounded-single-construction / independe	run; 0 exact closures and 0 mechanically admitted probes
20	reverse lexicon role noun boundary	reverse-lexicon-typed-complement-frame / semantic-role-lattice / object-noun-boundary-selection / outer-residual-reduction / heldout-scene / independent-po	repair run; 0 exact closures and 0 mechanically admitted probes
21	centerout grammar boundary repair5	center-out-grammar-state / residual-third-boundary-repair / heldout-scene-adjunct-authorship / event-preserving-valency / independent-exact-hash-audit	run; 0 exact closures and 0 mechanically admitted probes
22	reverse lexicon typed complement frame	reverse-lexicon-centered-complement / typed-transitive-complement-frame / live-reverse-chart-argument-selection / inherited-singular-agreement / heldout-sc	repair run; 0 exact closures and 0 mechanically admitted probes
23	centerout grammar boundary repair4	center-out-grammar-state / next-character-obligation-repair / heldout-semantic-adjunct-authorship / event-preserving-valency / independent-exact-hash-audit	run; 0 exact closures and 0 mechanically admitted probes
24	reverse lexicon centered complement	reverse-lexicon-shared-agreement-seams / centered-complement-clause / inherited-subject-number / three-live-obligations / heldout-scene / independent-point	repair run; 0 exact closures and 0 mechanically admitted probes
25	centerout grammar boundary repair3	center-out-grammar-state / two-boundary-obligation-repair / heldout-time-adjunct-authorship / event-preserving-valency / independent-exact-hash-audit	run; 0 exact closures and 0 mechanically admitted probes
26	reverse lexicon shared agreement seams	reverse-lexicon-boundary-seam-repair / two-lexical-boundaries / shared-singular-agreement-state / cross-seam-residual-closure / heldout-scene / independent	repair run; 0 exact closures and 0 mechanically admitted probes
27	centerout grammar boundary repair2	center-out-grammar-state / outer-letter-directed-adjunct-authorship / heldout-semantic-boundary-repair / event-preserving-valency / independent-exact-hash-	run; 0 exact closures and 0 mechanically admitted probes
28	reverse lexicon boundary seam repair	reverse-lexicon-inflection-repair / joint-determiner-adjective-seam / reverse-obligation-boundary-chart / heldout-scene / independent-pointer-sha-audit	repair run; 0 exact closures and 0 mechanically admitted probes
29	centerout grammar boundary repair	center-out-grammar-state / first-mismatch-boundary-repair / heldout-adjunct-substitution / event-preserving-valency / independent-exact-hash-audit	run; 0 exact closures and 0 mechanically admitted probes
30	reverse lexicon inflection repair	reverse-lexicon-synthesis / boundary-aware-inflectional-chart / live-suffix-obligation-selection / heldout-scene / independent-pointer-sha-audit	repair run; 0 exact closures and 0 mechanically admitted probes
31	centerout grammar boundary dp	center-out-grammar-state / lexical-boundary-dynamic-programming / bilateral-character-frontier / agreement-valency-state / whole-prose-rendering / independ	run; 0 exact closures and 0 mechanically admitted probes

#	Method / operator	Construction signature	Observed result and next repair
32	reverse lexicon synthesis	left-to-right-semantic-clause / reverse-lexicon-character-chart / nonpalindromic-word-realization / live-mirror-obligation / independent-pointer-sha-audit	run; 0 exact closures and 0 mechanically admitted probes
33	clause growth frame repair	authored-scene-clause-growth / typed-valency-frame-lattice / global-character-debt-ledger / heldout-semantic-frame-repair / ordinary-order-composition / in	run; 0 exact closures and 0 mechanically admitted probes
34	online grammar state char decoder	online-grammar-state / two-frontier-character-actions / live-obligation-ledger / word-boundary-language-score / ordinary-order-clause-state / independent-h	run; zero exact closures
35	online grammar state slot repair	online-grammar-state / heldout-first-mismatch-slot-repair / ledger-replay / semantic-role-preservation / independent-hash-audit	run; zero exact closures
36	online grammar state outer frame repair	online-grammar-state / authored-outer-edge-compatible-frame / live-ledger-replay / semantic-scene-realization / independent-hash-audit	run; zero exact closures
37	paraphrase graph debt paths	complete-authored-scene-graph / meaning-preserving-paraphrase-edges / two-normal-order-paths / mirrored-character-debt / heldout-edge-repair / independent-	run; 0 exact closures and 0 mechanically admitted probes
38	assumption core scene solver	assumption-literal-constraint-ledger / minimal-conflict-core / ordinary-order-scene-slots / joint-grammar-tense-agreement-equation / heldout-one-slot-repai	run; zero exact closures
39	induced pcfg character equation	induced-pcfg-from-attested-prose / probabilistic-derivation-sampling / bilateral-character-equation / semantic-role-unification / attested-pp-repair	run
40	voice alternation residual	voice-alternation-residual / authored-active-passive-clauses / agreement-aware-voice-switch / mirrored-character-ledger / independent-exact-audit / no-cata	run
41	dream rsi online round110	dream-rsi / round110 / online-redeployment / branching-two-region	9-node fresh tree; all controls non-exact; no reader gate
42	dream rsi masked scene repair	dream-rsi / masked-scene-repair / typed-slot-reopen / monotone-mismatch	1 monotone repair accepted; 0 exact closures; next repair jointly reopens mirrored verb-object slots
43	compositional grammar spine	compositional-grammar-spine / typed-valency-agreement / pre-render-character-equations / scene-first	36 fresh nodes; 0 exact closures; next repair permits cross-slot noun/adjunct character carry
44	dream rsi round111	dream-rsi / round111 / noun-number-masked-compositional / offline-replay	564 worlds, 21,729 audited nodes, 12 branching parents; 0 admissible exact rows
45	reciprocal plural frame seam	reciprocal-frame / plural-object-agreement / semantic-valency / pre-render-equation	0 equation-compatible nodes and 0 exact closures; next repair adds reciprocal preposition alternations
46	dream rsi round112	dream-rsi / round112 / reciprocal-plural-frame-seam / offline-replay	564 worlds and 21,729 audited nodes; 0 admissible exact rows
47	reciprocal prep typing seam	reciprocal-frame / preposition-alternation / animate-inanimate-typing / pre-render-equation	0 equation-compatible nodes and 0 exact closures; next repair allows typed determiner/adjunct attachment changes
48	dream rsi round113	dream-rsi / round113 / reciprocal-prep-typing-seam / offline-replay	564 worlds and 21,729 audited nodes; 0 admissible exact rows
49	typed determiner adjunct seam	typed-attachment / determiner-adjunct-geometry / reciprocal-valency / pre-render-equation	0 equation-compatible nodes and 0 exact closures; next: cross-attachment boundary carry
50	dream rsi round114	dream-rsi / round114 / typed-determiner-adjunct-seam / offline-replay	564 worlds and 21,729 audited nodes; 0 admissible exact rows

5. Reader-facing protocol and next constructive repair

No row has reached the reader study in this snapshot. When one does, the package will contain the intact passage, a word-shuffled control, randomized blinded order, the exact normalization and hash audit, provenance and generator commit, and a reproducible rater form. The immediate next experiment is not another sweep: jointly generate a complete question-answer discourse across the seam, carry agreement and valency state, and reject any candidate containing a proper palindromic subspan before it can be considered for readers.

Reproducibility anchors

Source artifacts: docs/experiment-novelty-registry.json; runs/dream-rsi-dual-boundary-model-authoring-20260918.json; runs/dream-rsi-discourse-ellipsis-20260918.json. The report is generated from these files so the table and headline counts can be regenerated after the next Dream-RSI branch.